Cubulating spaces and groups

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Contents

1 Basic notions 1

Lecture 1

1 Basic notions

An *n-cube* is $c = [0,1]^n$. A *face* of c is the restriction of some cooridnates to $\{0,1\}$.

A *cube complex* is a cell complex, built out of cuves. Let

- $\mathcal{C}$ be a disjoint union of cubes,
- $\mathcal{I}$ be a collection of isometries between faces of the same dimension.

Then the cell complex is $X = \mathcal{C}/\mathcal{I}$. The *dimension* of X is

$$\dim(X) = \sup\{\dim(c) \mid c \text{ is a cube in } X\}$$

We say that X is *finite* if it has finitely many cubes, and X is *locally finite* if each $p \in X$ lies in finitely many cubes.

We will make the assumption in this course that unless stated otherwise, cube complexes are locally finite.

The *link of a vertex in X* is, intuitively, the ε-sphere around $v \in X^{(0)}$. To make this formal, for an n-cube c, we define

$$\operatorname{link}_c(v) = S^{n-1}(v, \varepsilon) \subseteq c$$

where $\varepsilon = 1/4$. Using this,

$$\operatorname{link}(v) = \frac{\bigsqcup \operatorname{link}_c(v)}{\mathcal{I}'}$$

where $\mathcal{I}'$ is the restriction of $\mathcal{I}$ to the faces.

A *simplex complex* is *flat* if

1. it is a simplicial complex, That is, we want the 1-skeleton to not have loops or bigons.
2. $n+1$ vertices span a simplex if and only if they are pairwise adjacent.

A cube complex X is *non-positively curved (npc)* if $\operatorname{link}(v)$ is flat for each $v \in X^{(0)}$. As a counterexample, we can see that loops and bigons can appear as links.

On the other hand, if we have

then this is npc if the 3-cube is in the complex. Finally, the cube complex

is npc. A *CAT(0)* cube complex is a npc cube complex which is simply connected. In particular, being a CAT(0) cube complex is the same as being universal covers of npc cube complexes.

We will use the notation $\widetilde{X}$ for CAT(0) cube complexes.

Example 1.1

An n-cube is npc. Similarly, any graph is npc.

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Euclidean space $\mathbb{E}^n$, tesselated by unit cubes, is a CAT(0) cube complex. The link of each vertex corresponds to 2^n simplices, which give us an S^{n-1}.

Example 1.2

The n-torus T^n is a npc cube complex. For example, T^3 can be given by gluing opposite faces of a cube.

Remark 1.3. If X is 2-dimensional, we call it a square complex. A square complex X is nonpositively curved if and only if the girth of each link(v) is at least 4. Recall that the *girth* of a graph Γ is the length of the shortest cycle.

To see this, when $\dim(X) = 2$ we can't have a triangle as a link, since it would require a 3-cell.

Theorem 1.4. Every surface S not homeomorphic to S^2 or $\mathbb{RP}^2$ is homeomorphic to a npc cell complex.

Proof. We will prove it for S closed orientable. The general case is left as an exercise. For this, recall that if S has genus g, then it is given by gluing edges of a $4g$-gon. Subdivide the $4g$-gon as follows:

It remains to show that the cube complex we get is npc. The central vertex has link being a $4n$-cycle. The original vertex has link a $4n$-cycle, and the subdivided vertices have link being 4-cycles. □

Some topological constructions:

(a) Being npc is preserved under cubical subdivision, i.e. subdividing an n-cube into 2^n n-cubes.

(b) If $X_1, \ldots, X_n$ are npc, then so is $X_1 \times \cdots \times X_n$. We leave this as an exercise.

(c) If X is npc, then every covering space of X is npc.

(d) In general, subcomplexes of a npc cube complex X is not npc. However, if $\dim(X) \leq 2$ then any subcomplex is npc.